

A Soliton Cellular Automaton

Author: Daisuke Takahashi and Junkichi Satsuma • *Department of Applied Physics, Faculty of Engineering, University of Tokyo, Tokyo 113, Japan*

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ABSTRACT

We propose a 1+1 dimensional deterministic cellular automaton that exhibits exact soliton behavior. The system consists of an array of boxes where each box can contain at most one ball. The time evolution is governed by simple local rules or an equivalent carrier mechanism. We demonstrate that contiguous clusters of balls behave like solitons in the continuous Korteweg-de Vries (KdV) equation: their velocity is proportional to their length, and collisions between solitons of different sizes are completely elastic, preserving their individual shapes and resulting in exact phase shifts. Because larger solitons travel faster than smaller ones, an initial arbitrary configuration naturally sorts its constituent solitons by length over time.

1. Ultradiscretization and Model Definition

The Box-Ball System (BBS) can be derived through the ultradiscretization of the continuous Korteweg-de Vries (KdV) equation and Toda lattice. Ultradiscretization replaces algebraic operations with the tropical (max-plus) semiring:

$x \text{ (plus) } y = \max(x, y), \quad x \text{ (times) } y = x + y$

via the fundamental limit: $\lim_{\epsilon \rightarrow 0} \epsilon \cdot \ln(\exp(A/\epsilon) + \exp(B/\epsilon)) = \max(A, B)$.

The Carrier Mechanism: Consider a 1D array of boxes indexed by i in $\mathbb{Z}$, with u_i in $\{0, 1\}$.

At each time step $t \rightarrow t+1$, an imaginary carrier moves from left to right:

1. **Carrier begins with $C = 0$ balls.**
2. At box i : if $u_i = 1$, carrier picks up ball: $u_i \leftarrow 0, C \leftarrow C + 1$.
3. If $u_i = 0$ and $C > 0$, carrier deposits one ball: $u_i \leftarrow 1, C \leftarrow C - 1$.
4. **If $u_i = 0$ and $C = 0$, box remains empty.**

2. Soliton Properties and Exact Elastic Collisions

Velocity Law: A contiguous cluster of L balls (a soliton of size L) separated by empty space moves exactly L boxes to the right in each time step: $v(L) = L$.

Elastic Phase Shifts: When a faster soliton of length L_1 overtakes a slower soliton of length L_2 ($L_1 > L_2$):

- Both solitons fully recover their original lengths L_1 and L_2 after interaction.
- The faster soliton is shifted forward by $2 * L_2$ boxes.
- The slower soliton is shifted backward by $2 * L_2$ boxes.

Spatial Sorting Property: Because velocity is strictly monotonic in soliton length ($v(L_1) > v(L_2)$ iff $L_1 > L_2$), as $t \rightarrow \infty$:

- Smaller solitons trail on the left, while larger solitons advance to the right.
- Reading the lattice from left to right yields the solitons in strictly ascending order of length: $L_{\{1\}} \leq L_{\{2\}} \leq \dots \leq L_{\{n\}}$.

This establishes the Box-Ball System as an ultradiscrete physical sorting automaton.